INCH-POUND

A-A-59149
1 January 1998

COMMERCIAL ITEM DESCRIPTION

MONITOR, POWER LINE

The General Services Administration has authorized
the use of this commercial item description (CID) for all federal agencies.

l. SCOPE. This commercial item description covers an eight channel power line monitor. This CID is meant as a minimum requirement for the power line monitor in which only those manufacturers that meet or surpass the following requirements are qualified in accordance with this CID.

2. SALIENT CHARACTERISTICS.

2.1 Equipment covered by this specification may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall meet requirements of MIL-PRF-28800 Class 3.

2.2 Power Consumption. The equipment maximum power consumption of all components combined shall not exceed 300 VA.

2.3 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources.

2.4 Bid Samples. A quantity of two (2) each of the equipment offered are required to be delivered in accordance with directives provided in the contract.

2.5 Self Test. The equipment shall provide automatic self-tests to indicate whether the equipment is operating within the performance specification.

2.6 Modem Interface Requirements. The equipment modem interface shall have the capability to control each device function including all ranges, modes ,and levels. Primary power control and other non-measurement functions need not be controllable
through the modem interface. The modem shall provide at least 14.4 baud transfer rates.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: SA-ALC/LDAE, 308 Avionics Circle, Suite 2, Kelly AFB, TX 78241-5947.

AMSC N/A FSC 6625

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2.7 Communication Ports. The equipment shall provide an RS-232 interface that complies with EIA/TIA 574 for data transfer, Centronics interface complying with C332-44 Feb 1977, Rev A for printer outputs.

2.8 Dimensions. The equipment physical dimensions shall not exceed 20 inches in width, 12 inches in height, or 24 inches in depth.

2.9 Weight. The weight, including front cover and all accessories (excluding manual), shall not exceed 30 pounds.

2.10 Rack-Mount. Rack-mount capability shall be available as an option.

2.11 Protection. The equipment shall have a cover protecting front panel controls and display if they are an integral part of the equipment.

2.12 Accessory Storage. The equipment shall provide for stowage of all accessories by a pouch or container.

2.13 Year 2000 Compliant. The equipment shall accurately process date and time data from, into, and between the twentieth and twenty-first centuries, and the years 1999 and 2000 and leap year calculations. This includes, but is not limited to, calculating, comparing, and sequencing date and time data. The equipment shall also accurately process date and time data if such data is received or exchanged from other equipment and sources.

2.13.1 Date Ranges. The equipment shall correctly process data containing: dates before 1 January 2000 from the present until 1 January 2000; dates after 31 December 1999 from the present until 1 January 2000; dates on and after 1 January 2000 containing dates before 1 January 2000; dates on or after 1 January 2000 containing dates after 31 December 1999.

2.13.2 Computing Dates. The equipment shall handle the date 01/01/00 (1 January 2000) correctly; Recognize Year 2000 as a leap year; recognize 29 February 2000 as a valid date; recognize Julian Date 00060 as 29 February 2000 if Julian date format is provided or utilized; recognize Julian Date 00366 as 31 December 2000 if Julian date format is provided or utilized; perform arithmetic operations recognizing that the Year 2000 has 366 days; recognize that binary date 36584 as 29 February 2000, if binary date format is utilized; display, print, input and store dates in unambiguous format in the context of their usage.

2.14 Function Calculations. The equipment shall correctly calculate functions using date fields for non-date related functionality if provided or utilized and must be valid for all dates included in but not limited to archiving, naming conventions, security passwords, hashing, random number generators, and tape storage.

2.15 Performance Characteristics. The equipment shall be capable of measuring, displaying, and storing wave-forms within the minimum ranges, specifications, and accuracies described in this CID when calibrated in accordance with manufacturer's approved procedures and described under calibration requirements contained herein.

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2.15.1 Acquisition Channels. The equipment shall acquire data on 8 channels simultaneously and all performance requirements specified herein shall be accomplished for all channels.

2.15.2 Sample Rate The equipment shall provide a minimum sample rate of 128 samples/second for each channel for voltage and current acquisition.

2.15.3 Impulsive Transient Capture. The equipment shall provide a minimum sample rate of 2 millisec/sec and/or peak detect to capture impulsive voltage and current transients.

2.15.4 Operation Modes. The equipment shall acquire data in three modes: power quality; energy or power flow, and harmonics. The instrument may acquire data in these modes either simultaneously or individually but will not require the changing or addition on any module to perform these measurement modes.

2.15.5 Voltage Channels. The equipment shall have at least four independent voltage measurement channels.

2.15.5.1 Power Configuration Single Phase Systems. The equipment shall measure single and dual single phase AC systems both grounded and ungrounded.

2.15.6 Power Configuration Polyphase Systems. The equipment shall measure three phase WYE, and three phase DELTA polyphase AC systems both grounded and ungrounded.

2.15.6.1 Input Voltage Range. The equipment shall provide an AC voltage range of 10 volts rms (Vrms) to 600 Vrms over the frequency range of 45 to 415 Hertz (Hz).

2.15.6.2 Crest Factor. The equipment shall have a minimum crest factor of 2.

2.15.6.3 Voltage Measurement Accuracy. The equipment shall have a minimum measurement uncertainty for rms voltages of ±(1.0% of reading (rdg) + 0.5% of full scale (fs)) over the input voltage and frequency ranges.

2.15.6.4 Voltage Connections. The equipment shall have interconnecting cables and any additional accessories to connect the equipment in single phase or polyphase configurations.

2.15.6.5 Voltage Triggers. The equipment shall trigger on defined events from preset and /or automatic thresholds on all voltage channels. The equipment shall record all channels when the trigger or preset threshold event occurs and will continue recording until the trigger conditions end. The equipment may use data compression techniques to record long term events that exceed memory length.

2.15.6.6 Voltage Channel Input Impedance. The equipment input impedance shall be at least 1 MΩ.

2.15.6.7 Voltage Thresholds. The equipment shall provide adjustable or automatic threshold settings in at least 1V and/or 1% increments over the input voltage range.

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2.15.6.8 Thresholds Settings. The equipment may provide thresholds based on IEEE Std 1159-1995 categories and typical characteristics of power system electromagnetic phenomena for duration and voltage magnitude.

2.15.6.9 Extended Voltage Range. The equipment shall monitor voltage levels greater than 600 Vrms by the use of potential transformers (PTs). The equipment shall provide means to record the PT ratio and all measurements shall automatically be adjusted by the PT ratio factor.

2.15.7 Current Channels. The equipment shall have at least four independent current measurement channels with differential inputs and shall record rms current simultaneously with the voltage channels.

2.15.7.1 Input Current Range. The equipment shall provide current measurement of at least 1 Ampere rms (Arms) by the use of current transformer (CT) probes.

2.15.7.2 Current Measurement Accuracy. The equipment shall have a minimum measurement uncertainty for rms currents of ±(1.0% of rdg + 0.5% of fs) with CT probes attached.

2.15.7.3 Current Thresholds. The equipment shall provide adjustable or automatic threshold settings in at least 1A and/or 1% increments over the input current range.

2.15.7.4 Current Triggers. The equipment shall trigger on defined events from preset and/or automatic thresholds on all current channels. The equipment shall record all channels when the trigger or preset threshold event occurs and will continue recording until the trigger conditions end. The equipment may use data compression techniques to record long term events that exceed memory length.

2.15.7.5 Extended Current Range. The equipment shall have CT probes for each channel which provide measurement capability over the ranges of 0.5 to 3000 Arms in no more than three overlapping ranges.

2.15.7.6 Current Probe Bandwidth. The equipment current probe(s) shall have a bandwidth of 45 Hz to 3 KHz minimum.

2.15.8 Impulsive Transient Measurements. The equipment shall provide measurement of impulsive transients over the voltage range of 100 V to 1000 V peak and current range of 3 A to 1000 A.

2.15.8.1 Impulsive Transient Measurement Uncertainty. The equipment shall measure impulsive transient amplitude with a minimum uncertainty of ±(10% of rdg + 1.0% of fs) for half sine wave pulses ≤10μs.

2.15.8.2 Impulsive Transient Occurrence. The equipment shall indicate the time the impulsive transient occurred within 1 second.

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2.15.9 Measurements. The equipment shall provide the following measurements as defined by IEEE-STD-1159 over the ranges specified for spectral content, duration, and magnitude.

2.15.9.1 Power Quality Measurements. The equipment shall monitor, capture, and display the following electromagnetic phenomena or electromagnetic disturbances: Voltage impulsive transients; Oscillatory transients; Instantaneous, momentary, and temporary voltage and current sags and swells; Long duration interruptions of over-voltage, under-voltage, and sustained interruptions; and voltage imbalance.

2.15.9.2 Power, Power Factor, and Consumption Measurements. The equipment shall measure and/or calculate: power in W, KW, VA, and KVA; True and displacement power factor; reactive power; and energy in Watt-hours (WH) and KWH.

2.15.9.3 Harmonic Measurements. The equipment shall monitor, capture, and display harmonics up to the 50^{th} and total harmonic distortion.

2.15.9.4 Events Over Time Recording. The equipment shall provide a recording of events over time in the following categories: Power quality for rms voltage current; Power flow or energy for watts, volt-amperes, rms voltage, and rms current; and harmonics for voltage and current total harmonic distortion, rms voltage, and rms current.

2.15.9.4.1 Instantaneous Event Capture. The equipment shall monitor, capture, and display at least one cycle of voltage and current wave-forms for steady-state analysis and harmonic trending. The time interval at which these instantaneous event captures occur shall be selectable from 1 minute to 24 hours in 1 minute increments.

2.15.10 Non-interruptible Power Supply. The equipment shall have a rechargeable non-interruptible power supply that will provide one hour of monitoring operation when equipment power is lost and the power supply life shall be at least two years.

2.15.11 Display. The equipment shall have a display, either as an integral part of the equipment or external, providing representations of captured data in graphical and data formats.

2.15.11.1 Phasor Diagrams. The equipment shall provide and display phasor diagrams indicating phase relationships between voltage, current and power.

2.15.11.2 Magnification. The equipment shall provide magnification capability to magnify wave-forms events for accurate measurements.

2.15.11.3 Cursors. The equipment shall provide the capability of selecting events by movement of vertical and horizontal indicators to select areas of interest.

2.15.11.4 Data Analysis. The equipment shall provide an analysis capability to view recorded events by use of external software and/or internal routines.

2.15.11.5 Date and Time Stamp. The equipment shall provide he time and date of each event recorded.

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2.15.12 Memory. The equipment shall provide mass storage for reference wave-forms, setups, data files and image files of at least 1Mbyte.

2.15.12.1 File Formats. The equipment shall have the minimum capability of providing, storing, and transferring data in image formats compatible with Microsoft Windows operating system version 3.1 and higher.

2.15.12.2 Data Interchange. The equipment shall provide data interchange for all captured data in both graphic format and spreadsheet data compatible with Microsoft Windows operating system version 3.1 or higher applications.

2.15.12.3 Mass Storage Devices. The equipment may use a floppy disk drive, a hard disk drive, PCMCIA I/II/III flash memory card or other storage device with 1.4 Mbyte or greater capacity and compatible with Microsoft Windows operating system version 3.1 and higher.

2.15.12.4 Mass Storage Operational Restraints. The equipment shall have the ability to store, archive, retrieve, and perform any other storage operation over a temperature range of +10°C to +50°C.

2.15.12.5 Memory Data Transfer. The equipment shall provide the capability to save any wave-forms or setup data to any applicable reference memory location, to transfer setup information and/or wave-forms into a reference memory location via the RS-232 or modem interface, and to transfer setup and wave-form data from the floppy disk drive.

2.15.13 Autoset. The equipment shall provide an Autoset function that, when activated, will automatically configure the equipment to monitor voltage, current, and harmonics at predetermined thresholds dependent upon the voltage range and configuration of the system under test supplied by the operator.

2.15.14 Required Accessories. The following (minimum) accessories shall be required with each deliverable equipment.

2.15.14.1 Probes. The equipment shall have active or passive probes provided for all channels which allow the equipment to meet all requirements of this Commercial Item. The CT and PT probe ratio shall be automatically recognized and the scale factor of the equipment adjusted appropriately.

2.15.14.2 Standard Accessories. The standard accessories as supplied by the manufacturer shall be included.

2.15.14.3 Additional Accessories. The following items shall be required with each deliverable equipment if not provided as standard accessories.

2.15.14.3.1 Tutorial/User Reference Manual.

2.15.14.3.2 U.S. Power Cord.

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3. REGULATORY PROVISIONS.

3.1 Recovered Material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR) and Public Law 94-580.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Self Test. The equipment shall provide automatic self-tests in accordance with MIL-PRF-28800 paragraph 3.3.6 which shall be invoked by front panel controls and may be invoked by software command over a communication port.

4.3 Calibration. The equipment shall meet the requirements of MIL-PRF-28800F for calibration.

4.4 Warranty. The equipment shall offer, as a minimum, a three year warranty on parts and labor to repair these instruments, should they experience a fault which is not obviously due to abuse or irresponsible actions. A label shall be affixed to the equipment providing the shipping address for warranty service and the period of the warranty by date. A note shall be on the label stating, "Return shipping address must be enclosed."

5. PACKAGING.

5.1 Preservation, Packaging, Labeling, and Marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in their corresponding inch-pound units. A request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.2 Source of Documents

6.2.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.2.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

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6.2.3 IEEE Standards are available from the Institute of Electrical & Electronics Engineers, 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331.

6.2.4 EIA Standards are available from the Electronics Industries Association, 2500 Wilson Blvd, Arlington, VA 22201-3834.

6.3 National Stock Numbers (NSNs). The following is a list of NSNs which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

6625-01-295-9832	626A-PKG-KM	Dranetz
6625-01-343-2539	626G-PKG-KM	Dranetz
6625-01-269-5438	A-A-59149	Procurement Document

6.4 Part Identification Number (PIN). National Stock Numbers of products meeting the requirements of this CID will be assigned part number AA59149 in the Federal Cataloging System. This action is for government purposes and does not constitute a requirement for the contractor.

MILITARY INTERESTS:

Custodians:
Air Force - 99
Navy - SH
Army - MI

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

Preparing Activity:
Air Force - 82

DoD Project No. 6625-0845

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-59149	2. DOCUMENT DATE (YYMMDD) 98/01/01

3. DOCUMENT TITLE
MONITOR, POWER LINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME SA-ALC/LDAE	b. TELEPHONE *(Include Area Code)* (1) Commercial: (210) 925-6314 (2) AUTOVON: 945-6314
c. ADDRESS *(Include Zip Code)* 308 Avionics Circle, Suite 2 Kelly AFB, TX 78241-5947	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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